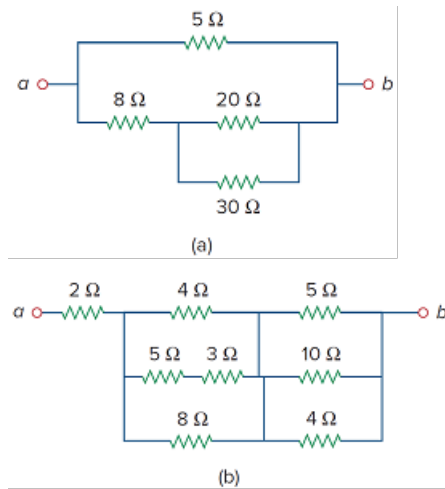


Problem

Reduce each of the circuits in Fig. 2.106 to a single resistor at terminals a - b .

Figure 2.106:



Step-by-step solution

Step 1 of 14

(a)

Consider the following circuit:

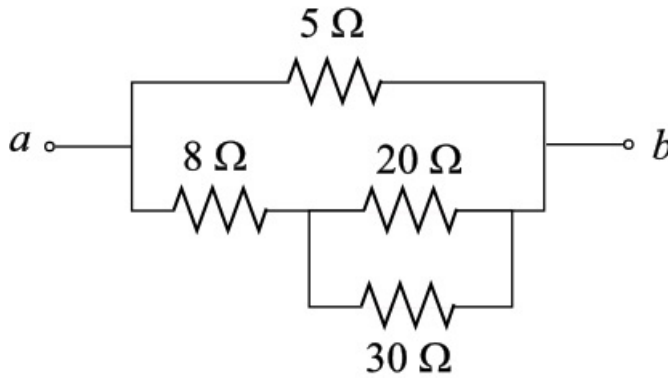


Figure 1

Step 2 of 14

In Figure 1, the $20\ \Omega$ resistor and the $30\ \Omega$ resistors are in parallel. Calculate the equivalent resistance.

$$\begin{aligned}
 R_{eq1} &= \frac{(20)(30)}{20 + 30} \\
 &= \frac{600}{50} \\
 &= 12\ \Omega
 \end{aligned}$$

Step 3 of 14

The equivalent circuit is shown in Figure 2.

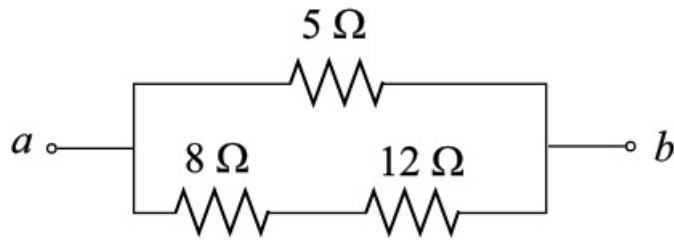


Figure 2

Step 4 of 14

From Figure 2, the equivalent resistance between terminals *a-b* is calculated as follows.

$$\begin{aligned} R_{a-b} &= 5 \parallel (8 + 12) \\ &= 5 \parallel 20 \\ &= \frac{(5)(20)}{5 + 20} \\ &= 4\ \Omega \end{aligned}$$

Step 5 of 14

Now the reduced circuit with a single resistor at terminals *a-b* is shown in Figure 3.

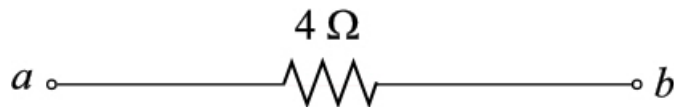


Figure 3

Step 6 of 14

(b)

Consider the following circuit:

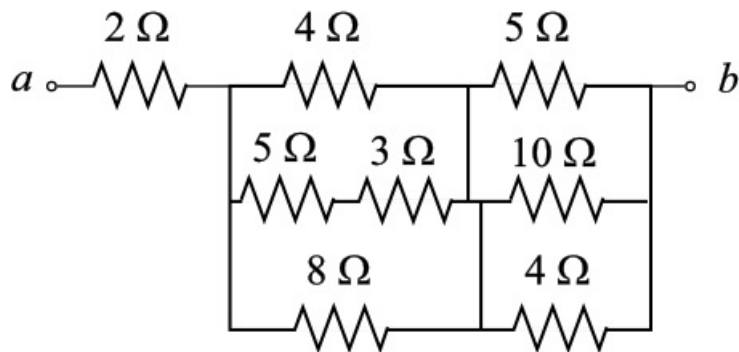


Figure 4

Step 7 of 14

In Figure 4, the $5\ \Omega$ resistor and the $3\ \Omega$ resistors are in series. Calculate the equivalent resistance.

$$\begin{aligned} R_{eq1} &= 5 + 3 \\ &= 8\ \Omega \end{aligned}$$

Step 8 of 14

The equivalent circuit is shown in Figure 5.

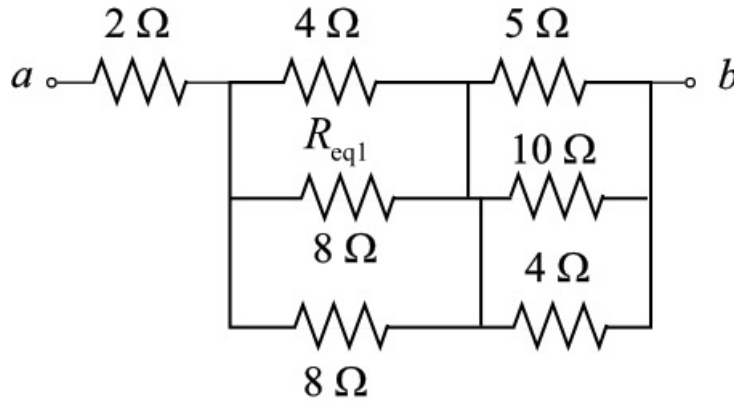


Figure 5

Step 9 of 14

In Figure 5, the two $8\ \Omega$ resistors and $4\ \Omega$ resistor are in parallel. Calculate the equivalent resistance.

$$\begin{aligned} \frac{1}{R_{eq2}} &= \frac{1}{8} + \frac{1}{8} + \frac{1}{4} \\ &= \frac{1+1+2}{8} \\ &= \frac{4}{8} \\ &= \frac{1}{2} \end{aligned}$$

Simplify further.

$$\begin{aligned} \frac{1}{R_{eq2}} &= \frac{1}{2} \\ R_{eq2} &= 2\ \Omega \end{aligned}$$

Step 10 of 14

The equivalent circuit is shown in Figure 6.

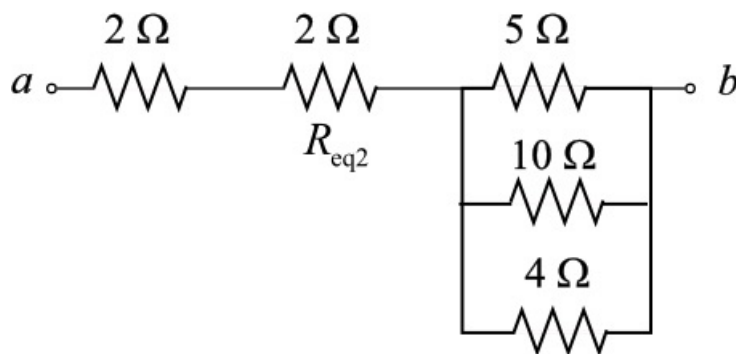


Figure 6

Step 11 of 14

In Figure 6, resistors, 4 Ω, 10 Ω, and 5 Ω are in parallel. Calculate the equivalent resistance.

$$\begin{aligned}\frac{1}{R_{eq3}} &= \frac{1}{4} + \frac{1}{10} + \frac{1}{5} \\ &= \frac{5+2+4}{20} \\ &= \frac{11}{20}\end{aligned}$$

Simplify further.

$$\begin{aligned}\frac{1}{R_{eq3}} &= \frac{11}{20} \\ R_{eq3} &= \frac{20}{11} \\ R_{eq3} &= 1.82 \, \Omega\end{aligned}$$

Step 12 of 14

The equivalent circuit is shown in Figure 7.

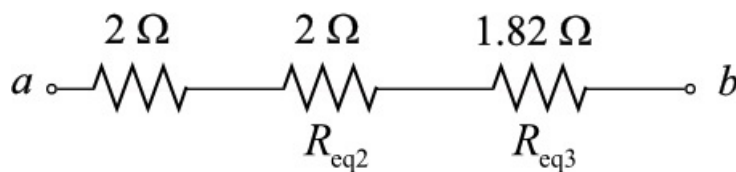


Figure 7

Step 13 of 14

In Figure 7, resistors, 2 Ω, 2 Ω, and 1.82 Ω are in series. Calculate the equivalent resistance.

$$\begin{aligned}R_{a-b} &= 2 + 2 + 1.82 \\ &= 5.82 \, \Omega\end{aligned}$$

Step 14 of 14

Now the reduced circuit with a single resistor at terminals a - b is shown in Figure 8.

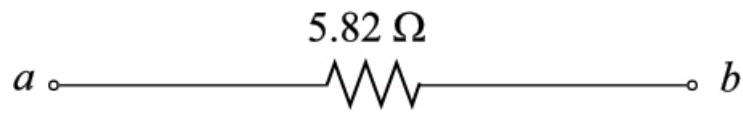


Figure 8